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(Test Case Design)

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Queueless - A Smart Online Queue Management System

Two features of this project are-

1. Digital Token Generation
2. Real-Time Queue Position Tracking

Digital Token Generation(EP)

EQUIVALENCE PARTITIONING

Partition	Description	Example
P1	Valid available service	General Service
P2	Invalid/non-existing service	Service-999
P3	Existing but unavailable service	Closed Service
P4	No service selected	Blank

EP TEST CASE:

TC ID	Input/Condition	Partition	Expected Result
EP-TG-05	Active session	P1 — Valid	Allow token generation
EP-TG-06	Guest user	P2 — Invalid	Ask user to log in
EP-TG-07	Expired session	P3 — Invalid	Ask user to log in again

TEST CASE DESIGN:

Field	Details
Test Case ID	TC-QUE-TG-01
Test Scenario	Verify that a logged-in customer can successfully generate a digital token for an available service.
Preconditions	1. Customer account is active. 2. Customer is logged in. 3. A valid service is available. 4. The selected queue has available capacity. 5. Customer does not already have an active token for the selected queue.
Test Steps	1. Log in to Queueless. 2. Navigate to the service/queue selection page. 3. Select an available service. 4. Click **Generate Token**. 5. Observe the generated token.
Input Data	Customer: Valid logged-in customer Service: General Service Queue: Available
Expected Result	The system successfully generates a unique digital token, displays the token number to the customer, and adds the token to the selected queue.
Actual Result	To be completed during test execution.
Status	Pass / Fail
Remarks	Validates the successful Digital Token Generation process.

Real-Time Queue Position Tracking (BVA)

The maximum active queue position is 100.

Test Value	Meaning
1	Minimum valid position
2	Just above minimum
99	Just below maximum
100	Maximum valid position
101	Above maximum

TEST CASES:

TC ID	Queue Position	Expected Result
BVA-QP-01	1	Position displayed correctly
BVA-QP-02	2	Position displayed correctly
BVA-QP-03	99	Position displayed correctly
BVA-QP-04	100	Position displayed correctly
BVA-QP-05	101	Invalid/out-of-range position handled

TEST CASE DESIGN:

Field	Details
Test Case ID	TC-QUE-QP-01
Test Scenario	Verify that the system displays the customer's current queue position in real time.
Preconditions	1. Customer is logged in. 2. Customer has a valid active token. 3. Customer is waiting in the queue. 4. Other customers are ahead of the customer.
Test Steps	1. Generate a token. 2. Join the selected queue. 3. Open Queue Position Tracking. 4. Observe the current queue position. 5. Serve one customer ahead of the current customer. 6. Observe the queue position again.
Input Data	Customer Token: T-025 Initial Position: 7 Customers served ahead: 1
Expected Result	The system initially displays position 7. After one customer ahead is served, the system updates the position to 6 without requiring an incorrect/manual queue calculation.
Actual Result	To be completed during test execution.
Status	Pass / Fail
Remarks	Validates real-time queue position tracking.